

Fixing KVM Acceleration Error in GNS3 on VirtualBox

Problem

While running GNS3 inside VirtualBox, the following error appears:

```
KVM acceleration cannot be used (/dev/kvm doesn't exist)
```

This issue occurs because nested virtualization is unavailable or disabled inside the VirtualBox VM.

Environment

Host OS	Windows 11
Virtualization Software	VirtualBox
Network Simulator	GNS3 VM 2.2.58.1
Emulator	QEMU (used inside GNS3)

Solution Used

Disabled KVM acceleration inside the GNS3 server configuration.

Steps Performed

Step 1 — Open the GNS3 VM

Start the GNS3 VM inside VirtualBox.

Step 2 — Open the GNS3 Configuration Menu

Use the following keyboard shortcut inside the VM:

```
Ctrl + Alt + T
```

Then select:

```
Configure
```

Description: Edit server configuration (advanced users ONLY)

Step 3 — Modify QEMU Settings

The [Qemu] section did not exist in the configuration file, so it was manually added. Add the following configuration:

```
[Qemu]
enable_kvm = false
```

This disables KVM acceleration for QEMU.

Step 4 — Save Configuration

Save the configuration and exit the editor.

Step 5 — Reboot the VM

Select "Reboot" from the GNS3 VM menu.

Result

GNS3 started successfully without the KVM acceleration error. QEMU now runs without hardware acceleration, which may reduce performance slightly but resolves compatibility issues when running GNS3 inside VirtualBox.

Optional: Better Solution (Enable Nested Virtualization)

For better performance, enable Nested VT-x/AMD-V in VirtualBox to allow KVM acceleration if supported by the host machine:

1	Power off the VM
2	Open VirtualBox Settings
3	Navigate to: System → Processor
4	Enable: Enable Nested VT-x/AMD-V

Note: This option may be greyed out on some systems if the CPU does not support nested virtualization or if Hyper-V is active on the Windows host.